

$$\begin{aligned}
 & \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) = \frac{1}{4} (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\
 & - (\langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle)) = \frac{1}{4} (2\langle u, v \rangle + 2\langle v, u \rangle) \\
 & = \frac{1}{4} (2(\langle u, v \rangle + \overline{\langle u, v \rangle})) = \frac{1}{4} (2 \cdot 2 \operatorname{Re}(\langle u, v \rangle)) = \\
 & = \underline{\operatorname{Re}(\langle u, v \rangle)}
 \end{aligned}$$

$$\begin{aligned}
 & \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2) = \\
 & = \frac{1}{4} (\langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle + \langle iv, iv \rangle - (\langle u, u \rangle + \langle u, iv \rangle + \langle iv, u \rangle + \langle iv, iv \rangle)) \\
 & = \frac{1}{4} (-2\langle u, iv \rangle - 2\langle iv, u \rangle) = \frac{1}{4} (-2(\langle u, iv \rangle + \overline{\langle u, iv \rangle})) \\
 & = \frac{1}{4} (-2(i \langle u, v \rangle - i \overline{\langle u, v \rangle})) = \frac{1}{4} (-2i \langle u, v \rangle + 2i \overline{\langle u, v \rangle}) \\
 & = \frac{1}{4} (-2i \langle u, v \rangle + 2i \overline{\langle u, v \rangle}) = \frac{1}{4} (-2i \langle u, v \rangle + 2i \overline{\langle u, v \rangle}) \\
 & = \frac{1}{4} (-4i \operatorname{Im}(\langle u, v \rangle)) = \underline{\operatorname{Im}(\langle u, v \rangle)}
 \end{aligned}$$